

Finite and Compact-Open Sets Are Sets of Synthesis for Abelian Locally Elliptic Weighted Orlicz Algebras

Automated Mathematics Run `fa_banach_001`, lane 2

21 July 2026

Abstract

Question 8.12 of Carter’s paper asks for geometric conditions on subsets of the primitive spectrum which imply spectral synthesis for Hermitian weighted Orlicz algebras on unipotent p -adic groups. We prove a complete criterion in the abelian locally elliptic case: every finite set and every compact-open set in the Pontryagin dual is a set of synthesis. In particular, this applies to the additive unipotent group $U_2(k) \cong (k, +)$ over every non-archimedean local field k . The mechanism is an exact compact-open Fourier cutoff which belongs to Carter’s minimal ideal $j(C)$.

Status. Candidate partial result, likely valid, subject to human review. The theorem fully proves the stated synthesis criteria in the abelian case; it does not resolve the nonabelian coadjoint-orbit problem.

1 Source question

The source is Max Carter, *Weighted Orlicz *-algebras on locally elliptic groups*, arXiv:2503.20735v2, Question 8.12 on page 29 [1]. Its core synthesis question is:

“What geometric properties of C imply that it is a set of synthesis?”

The surrounding source passage and the complete question are shown below.

2 Definitions and scope

Let G be an abelian locally elliptic group. Let (Φ, Ψ) be a complementary pair of Young functions with $\Phi \in \Delta_2$, and let ω be an L^Φ -weight such that $1/\omega \in L^\Psi(G)$. Put

$$\mathcal{A} = L^\Phi(G, \omega),$$

and assume that $\mathcal{A}$ is Hermitian. Carter’s Theorem 6.2 identifies $\text{Prim}_*(\mathcal{A})$ with the unitary dual; because G is abelian, this is the Pontryagin dual $\widehat{G}$.

For closed $C \subseteq \widehat{G}$, write

$$\ker(C) = \{f \in \mathcal{A} : \widehat{f}(\chi) = 0 \text{ for all } \chi \in C\}.$$

Carter defines a closed ideal $j(C)$ using a smooth functional calculus and proves $j(C) \subseteq \ker(C)$ and $\text{hull}_*(j(C)) = C$ [1, Theorem 7.1]. The set C is a *set of synthesis* when $j(C) = \ker(C)$.

Let $\mathcal{D}(G)$ be the span of the indicators of compact-open cosets of G . For the Fourier transform we use

$$\widehat{f}(\chi) = \int_G f(x) \overline{\chi(x)} dx.$$

Hermitian (which is the case if ω is sub-additive, for example) and suppose that I is a closed ideal of $L^\Phi(N, \omega)$ with $\text{hull}_*(I) := C \subseteq \text{Prim}_*(L^\Phi(N, \omega))$. Let $j(C)$ be the ideal of $L^\Phi(N, \omega)$ as defined in Theorem 7.1. Then, the ideal I must sit between $\ker(C)$ and $j(C)$ i.e. $j(C) \subseteq I \subseteq \ker(C)$. In particular, understanding the closed ideals of $L^\Phi(G, \omega)$ with hull equal to C would require one to understand the ideal theory of the algebra $\ker(C)/j(C)$.

The set $C \subseteq \text{Prim}_*(L^\Phi(N, \omega))$ is called a **set of synthesis** if $\ker(C)/j(C)$ is trivial, or equivalently, $\ker(C)$ is the only closed ideal of $L^\Phi(N, \omega)$ with hull C . The property of $L^\Phi(N, \omega)$ being Wiener is equivalent to the empty set being a set of synthesis, so under our assumption that $L^\Phi(N, \omega)$ is Hermitian, it follows that the empty set is a set of synthesis by Theorem 1.3. Also, by the previous corollary, we may identify C with a subset of $\mathfrak{n}^*/\text{Ad}^*(N)$. We thus pose the following problem.

Question 8.12. *Let N be a group of n -dimensional unipotent matrices over a p -adic field. Let (Φ, Ψ) be a complementary pair of Young functions and ω an L^Φ -weight on N with $1/\omega \in L^\Psi(N)$ and $L^\Phi(N, \omega)$ Hermitian. Given*

$$C \subseteq \mathfrak{n}^*/\text{Ad}^*(N) \cong \text{Prim}_*(L^\Phi(N, \omega)),$$

can geometric properties of C , when viewed as a subset of $\mathfrak{n}^/\text{Ad}^*(N)$, be linked to algebraic properties of the algebra $\ker(C)/j(C)$? In particular, what geometric properties of C imply that it is a set of synthesis?*

Figure 1: Question 8.12 and its definition of a set of synthesis, rendered from page 29 of arXiv:2503.20735v2.

3 Proof intuition

The dual of an abelian locally elliptic group has a compact-open basis. On such groups, compactly supported locally constant functions are Fourier stable. If $U \subseteq \widehat{G}$ is compact-open and disjoint from C , the inverse Fourier transform e_U of 1_U is therefore an honest element of $\mathcal{D}(G) \subseteq \mathcal{A}$.

The key point is stronger than ordinary Fourier localization: e_U lies in Carter's specific ideal $j(C)$. Scale e_U to a self-adjoint h whose Fourier transform takes only the values 0 and $a > 0$, and apply Carter's smooth functional calculus with a function that is zero near 0 and equal to 1 at a . The resulting element is exactly e_U and belongs to the generating set $m(C)$ of $j(C)$. Density then reduces synthesis to arranging that approximants have Fourier support away from C . For finite C this is done by finite interpolation; for compact-open C it is done by the Fourier projection 1_C .

4 Formal proof

Lemma 1 (Fourier test functions and continuity). *The following hold.*

- (i) $\mathcal{D}(G)$ is dense in $\mathcal{A}$.
- (ii) Fourier transformation maps $\mathcal{D}(G)$ onto $\mathcal{D}(\widehat{G})$.
- (iii) For every $\chi \in \widehat{G}$, evaluation $f \mapsto \widehat{f}(\chi)$ is continuous on $\mathcal{A}$.

Proof. Since $\Phi \in \Delta_2$, compactly supported functions are dense in the weighted Orlicz space; this is the standard Δ_2 density property [2]. A locally elliptic group has a compact-open basis. Hence a continuous function on a compact set is uniformly approximable by locally constant step functions

supported on finitely many compact-open cosets. On a fixed compact set K , uniform error ε gives Orlicz error at most $\varepsilon\|\omega 1_K\|_{L^\Phi}$ (up to the fixed equivalent norm convention). This proves (i).

If $K \leq G$ is compact-open and $x \in G$, Haar orthogonality gives

$$\widehat{1_{x+K}}(\chi) = \mu(K)\overline{\chi(x)}1_{K^\perp}(\chi).$$

Here $K^\perp$ is compact-open in $\widehat{G}$. The same formula on $\widehat{G}$, together with Fourier inversion, proves (ii); see also [3].

Finally, the Orlicz–Hölder inequality and $1/\omega \in L^\Psi(G)$ give a constant H such that

$$|\widehat{f}(\chi)| \leq \|f\|_1 \leq H\|f\|_{L^\Phi(G,\omega)},$$

which proves (iii). $\square$

Lemma 2 (Exact cutoffs lie in $j(C)$). *Let $C \subseteq \widehat{G}$ be closed and let $U \subseteq \widehat{G}$ be compact-open with $U \cap C = \emptyset$. If $e_U \in \mathcal{D}(G)$ is defined by $\widehat{e_U} = 1_U$, then $e_U \in j(C)$. Consequently, every $g \in \mathcal{D}(G)$ satisfying $\text{supp } \widehat{g} \subseteq U$ belongs to $j(C)$.*

Proof. The function e_U is self-adjoint because its Fourier transform is real. Choose $a \in (0, 2\pi)$ so small that $a\|e_U\|_1 \leq 1$, and set $h = ae_U$. Then $h = h^* \in C_c(G)$, $\|h\|_1 \leq 1$, and

$$\widehat{h} = a1_U.$$

In particular, every representation indexed by C annihilates h .

Fix the smooth functional-calculus algebra A_γ used in Carter’s definition of $m(C)$, with $\log_2(4/3) < \gamma < 1$. Its compact cutoff property supplies $\varphi \in A_\gamma$ which is zero on a neighbourhood of 0 and satisfies $\varphi(a) = 1$ [1, Proposition 5.5]. Thus $\varphi\{h\} \in m(C)$. At every $\chi \in \widehat{G}$, compatibility of the functional calculus with representations gives

$$\widehat{\varphi\{h\}}(\chi) = \varphi(\widehat{h}(\chi)) = 1_U(\chi).$$

Fourier injectivity on $L^1(G)$ yields $\varphi\{h\} = e_U$. Therefore $e_U \in m(C) \subseteq j(C)$. If $\text{supp } \widehat{g} \subseteq U$, then $g = g * e_U$, so the ideal property gives $g \in j(C)$. $\square$

Theorem 3 (Finite and compact-open synthesis). *Under the assumptions above:*

- (a) *every finite subset of $\widehat{G}$ is a set of synthesis for $\mathcal{A}$;*
- (b) *every compact-open subset of $\widehat{G}$ is a set of synthesis for $\mathcal{A}$.*

Proof. First let $F = \{\chi_1, \dots, \chi_r\} \subseteq \widehat{G}$. Choose pairwise disjoint compact-open neighbourhoods V_i of the χ_i , and let $q_i \in \mathcal{D}(G)$ satisfy $\widehat{q_i} = 1_{V_i}$. Thus $\widehat{q_i}(\chi_j) = \delta_{ij}$.

Take $f \in \ker(F)$ and choose $f_n \in \mathcal{D}(G)$ with $f_n \rightarrow f$ in $\mathcal{A}$. Define

$$g_n = f_n - \sum_{i=1}^r \widehat{f_n}(\chi_i)q_i.$$

By Lemma 1(iii), $\widehat{f_n}(\chi_i) \rightarrow \widehat{f}(\chi_i) = 0$, so $g_n \rightarrow f$ in $\mathcal{A}$. Moreover $\widehat{g_n}$ vanishes at every point of F . Since $\widehat{g_n}$ is locally constant, it vanishes on a neighbourhood of each such point. Hence $U_n = \text{supp } \widehat{g_n}$ is compact-open and disjoint from F . Lemma 2 gives $g_n \in j(F)$. Closedness gives $f \in j(F)$. Thus $\ker(F) \subseteq j(F)$; the reverse inclusion follows from the definition of $j(F)$. This proves (a).

Now let $C \subseteq \widehat{G}$ be compact-open and choose $q_C \in \mathcal{D}(G)$ with $\widehat{q}_C = 1_C$. If $f \in \ker(C)$, then $f * q_C = 0$ by Fourier injectivity. For $f_n \in \mathcal{D}(G)$ converging to f , put

$$g_n = f_n - f_n * q_C.$$

The Banach algebra inequality gives

$$\|g_n - f\|_{\mathcal{A}} \leq \|f_n - f\|_{\mathcal{A}} + \|(f_n - f) * q_C\|_{\mathcal{A}} \leq (1 + \|q_C\|_{\mathcal{A}})\|f_n - f\|_{\mathcal{A}},$$

so $g_n \rightarrow f$. Also $\widehat{g}_n = \widehat{f}_n(1 - 1_C)$; hence its compact-open support is disjoint from C . Lemma 2 gives $g_n \in j(C)$, and closedness again gives $f \in j(C)$. Therefore $\ker(C) = j(C)$, proving (b). $\square$

Corollary 4 (Additive p -adic unipotent case). *Let k be a non-archimedean local field and $N = U_2(k) \cong (k, +)$. Under Carter’s hypotheses $\Phi \in \Delta_2$, $1/\omega \in L^\Psi(N)$, and Hermitianness of $L^\Phi(N, \omega)$, every finite subset and every compact-open subset of*

$$\mathfrak{n}^* / \text{Ad}^*(N) = \mathfrak{n}^* \cong \text{Prim}_*(L^\Phi(N, \omega))$$

is a set of synthesis.

Proof. The group $U_2(k)$ is the additive group of k , hence abelian and locally elliptic. Its coadjoint action is trivial, so Theorem 3 applies. $\square$

5 Verification, limitations, and novelty check

The proof is structural and uses no numerical computation. The independent verification checklist confirms: (1) the Δ_2 hypothesis supplies the required dense Fourier test algebra; (2) the cutoff element satisfies every condition in the source definition of $m(C)$; (3) finite interpolation coefficients tend to zero by the continuous embedding $\mathcal{A} \hookrightarrow L^1(G)$; and (4) the compact-open projection preserves convergence in the Banach algebra norm.

The theorem does not address nonabelian N , nontrivial coadjoint orbit spaces, or arbitrary closed subsets of $\widehat{G}$. A bounded novelty search on 21 July 2026 covered arXiv queries combining “weighted Orlicz”, “spectral synthesis”, “ p -adic”, “finite sets”, and “compact open sets”, plus the source’s cited discussion. It found the source paper but no exact prior statement of this weighted-Orlicz criterion. This is not an exhaustive literature claim, so novelty confidence is moderate.

Human-review recommendation. Review the density step for the author’s chosen Orlicz norm convention and check that Proposition 5.5 supplies a periodic cutoff with $\varphi(a) = 1$ for the selected $a \in (0, 2\pi)$. The remaining argument is an exact Fourier-idempotent calculation.

References

- [1] M. Carter, *Weighted Orlicz *-algebras on locally elliptic groups*, *Studia Math.* **287** (2026), no. 2, 145–180; arXiv:2503.20735v2.
- [2] M. M. Rao and Z. D. Ren, *Theory of Orlicz Spaces*, *Monographs and Textbooks in Pure and Applied Mathematics* 146, Marcel Dekker, 1991.
- [3] W. Rudin, *Fourier Analysis on Groups*, *Interscience Tracts in Pure and Applied Mathematics* 12, Wiley, 1962.